ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** the container and the flame also differ along with the volume	Volume of water is a variable.
2	**(a)** both depend on hot weather	A third factor links them.
3	**(b)** compare the methods and repeat carefully	Differences in method may explain.
4	**(a)** more than one variable was changed	Only one variable should vary.
5	**(d)** amount of water	Water amount varies.
6	**(b)** rounding off numbers to look neat without noting it	Data must not be altered.
7	**(d)** revise the hypothesis	Data must be respected.
8	**(a)** growth of the plant	Growth is the dependent variable.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A hypothesis is a testable explanation, and testing it by an experiment is exactly how we check it.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the use of a telescope does not explain why observation comes first.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two factors differ: light and water (1); We cannot tell which factor caused the difference (1).	1+1
12	It represents the Earth in small size (1); It helps us study things too large to observe wholly (1).	1+1
13	Example: plastic waste (1); Science develops recycling or biodegradable materials (1).	1+1
14	It shows what happens without the change (1); Example: unfertilised plants compared to fertilised ones (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use same size, same wetness (1); Same place, time and temperature (1); Measure drying time for each (1).	1+1+1
16	Curiosity: why do things fall (1); Observation: fall of apple and Moon's motion (1); Reasoning: same force acts on both (1).	1+1+1
17	A factor that can change (1); Temperature of water (1); Amount of sugar and stirring (1).	1+1+1
	OR: P vs Q tests light (1); P vs R tests water (1); Only one factor differs in each comparison (1).	1+1+1
18	Hand is unreliable (1); It depends on earlier experience (1); A thermometer gives exact readings (1).	1+1+1
	OR: New experiments contradict it (1); A better explanation covers more facts (1); Example: Earth at centre idea was given up (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A hypothesis (1). (ii) The amount of sunlight (1). (iii) Recording the observations (or results) (1). (iv) Only one thing should change so that any difference can be blamed on sunlight (1); the plant in sunlight grew better, so plants need sunlight (1).	1+1+1+2
20	Sum = 90 s (1); Average = 30 s (1); One swing = 1.5 s (1); Timing many swings reduces error in starting and stopping (1); Dividing gives the time for one swing (1).	1+1+1+1+1
	OR: Parallax error (1); Wrong zero mark (1); Using broken scale (1); Reading from the correct angle (1); Repeating and averaging (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a guess a **conclusion**: the guess before the experiment is a hypothesis; the conclusion comes after the results.
- Changing many things in an experiment at once; only **one** condition should be changed.
- Writing only what you expected to see instead of what you actually observed.

- Believing that one trial is enough; experiments are repeated to be sure.
- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** curiosity about everyday things can lead to great discoveries	Simple tools with careful thinking can lead to discoveries.
2	**(b)** the plant grows with time	Rising line indicates growth.
3	**(d)** the half field not sprayed	The control is the unchanged group.
4	**(d)** it follows from the data collected under fair conditions	Validity depends on evidence.
5	**(d)** they can measure precisely and consistently	Precision and consistency.
6	**(c)** repeated measurements by different people	Repeated, independent evidence is strong.
7	**(c)** The rod is 25 cm long	It involves a number.
8	**(c)** verification	Others can verify.
9	**(c)** A is true but R is false.	A is true because repeating avoids mistakes; R is false, a single trial can be wrong.
10	**(d)** A is false but R is true.	A is false: science keeps growing and old ideas may be improved. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One case may be a coincidence (1); More cases are needed to see a pattern (1).	1+1
12	It is a single case (1); He may have recovered naturally; trials are needed (1).	1+1
13	Average = 39 / 3 = 13 s (1); It reduces the effect of random errors (1).	1+1
14	A very different value stands out (1); It can be examined and the cause found (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Two things may rise together (1); Example: ice-cream sales and sunburn (1); Both depend on a third factor, hot weather (1).	1+1+1
16	It can be tested with evidence (1); Scientific: does heat affect melting of wax (1); Non-scientific: which flower is lovelier (1).	1+1+1
17	Eyes see colour and shape (1); Nose notes smell (1); Magnifying lens shows the tiny parts (1).	1+1+1
	OR: Different measuring habits (1); Different conditions such as temperature (1); Errors in instruments or reading (1).	1+1+1
18	They used observation and reasoning (1); Their records influenced later knowledge (1); They inspire scientific attitude (1).	1+1+1
	OR: Honest data builds trust (1); Teams share work, such as large experiments (1); Different skills solve complex problems (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Questioning or checking claims (curiosity) (1). (ii) Observe carefully (1). (iii) Records stop us forgetting and let us compare and study the data later (1). (iv) Many observations show whether the pattern repeats, so the conclusion is more reliable (1) and a chance event on one day does not mislead us (1).	1+1+1+2
20	Telescope revealed moons and stars (1); Microscope revealed cells (1); Earlier beliefs changed (1); New questions arose (1); New branches of science developed (1).	1+1+1+1+1
	OR: Science answers testable questions (1); Many questions remain unanswered (1); New discoveries pose new questions (1); Values and opinions are outside science (1); Knowledge keeps growing (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a guess a **conclusion**: the guess before the experiment is a hypothesis; the conclusion comes after the results.
- Changing many things in an experiment at once; only **one** condition should be changed.

- Writing only what you expected to see instead of what you actually observed.
- Believing that one trial is enough; experiments are repeated to be sure.
- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **The Wonderful World of Science** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** It can have both benefits and harmful effects	Responsible use is important.
2	**(d)** question, hypothesis, test, result, conclusion, sharing	This is the logical sequence.
3	**(c)** new evidence can change them	They may be revised.
4	**(d)** cells and microbes	Microscopes revealed the unseen world of organisms.
5	**(d)** to represent things too large or small to handle, like the solar system	Models help to visualise.
6	**(c)** others can get the same result	Replication builds trust.
7	**(b)** measuring your height	Simple measurement needs no hypothesis.
8	**(a)** it can only answer questions that can be tested	Science studies testable questions.
9	**(a)** Both A and R are true and R is the correct explanation of A.	Because it magnifies, tiny things become visible; R explains A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but one does not explain the other.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Others check methods and results (1); It catches errors and improves quality (1).	1+1
12	It is an opinion that cannot be tested (1); There is no measurable evidence (1).	1+1
13	It would be a guess rather than a result (1); Only what the evidence supports can be claimed (1).	1+1
14	A common system avoids confusion (1); Results can be compared everywhere (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	A metal rod of same length after heating (1); If length stays the same (1); Repeated trials show no change (1).	1+1+1
16	Table organises readings (1); Graph shows pattern or trend (1); Both make comparison easy (1).	1+1+1
17	Measure mass before soaking (1); Measure mass after soaking (1); Difference is the water absorbed (1).	1+1+1
	OR: Water purifiers (1); Rain water harvesting (1); Drip irrigation to save water (1).	1+1+1
18	Check for errors (1); Repeat the measurement (1); Not delete it without reason (1).	1+1+1
	OR: Dirt can change results (1); Old chemicals can react (1); It prevents contamination (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Microscope (1). (ii) Telescope (1). (iii) Astronomy (1). (iv) Our senses cannot see very tiny or very distant things (1); instruments also give more exact and reliable measurements (1).	1+1+1+2
20	Question and observation (1); Hypothesis (1); Tests and data (1); Sharing and review (1); Repeated confirmation (1).	1+1+1+1+1
	OR: Change only light level (1); Control: same seeds, soil, water (1); Measure height daily with a ruler (1); Record in a table (1); Handle soil and tools safely (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a guess a **conclusion**: the guess before the experiment is a hypothesis; the conclusion comes after the results.
- Changing many things in an experiment at once; only **one** condition should be changed.
- Writing only what you expected to see instead of what you actually observed.
- Believing that one trial is enough; experiments are repeated to be sure.
- Mixing up the instruments: microscope is for tiny things, telescope is for faraway things.
- Saying science facts never change; new evidence can improve our understanding.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND